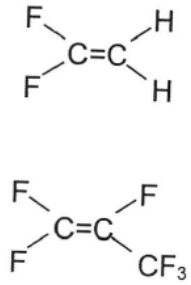
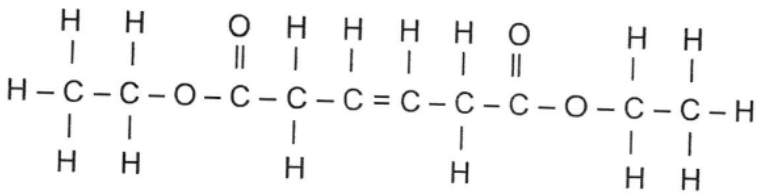
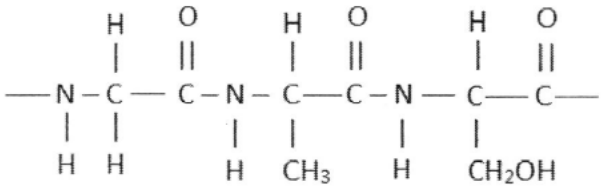
Answers

Organic Chemistry Test 3.0

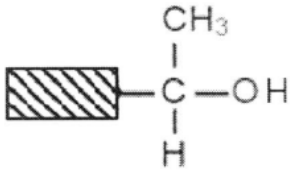
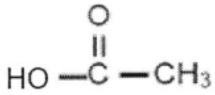
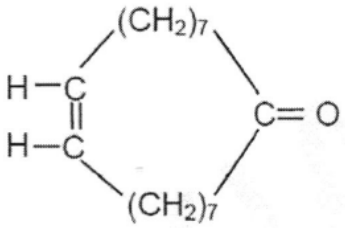
Q1

(a)	(i)	Addition polymerization	1
	(ii)		1 1
	(iii)	Mr of repeating unit = 214 No. of repeating units = 500 000 ÷ 214 = 2336 No. of monomers = 2336 × 2 = 4672 (or 4673)	1 1
(b)	(i)	Aq bromine: Reddish brown aqueous bromine is rapidly decolourised/turned colourless. Aq sodium carbonate: Effervescence occurs/ colourless gas produced/ bubbling.	1 1
	(ii)		1: 1 ethyl on left 1: 1 ethyl on right
	(iii)	Warming with concentrated sulfuric acid	1

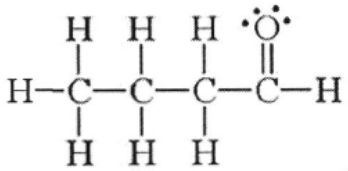
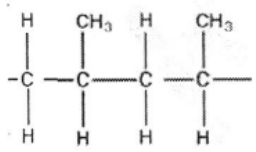
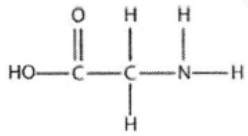
Q2

(a)(i)	$2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$;	[1]
(a)(iii)	Less intermolecular forces of attraction between isooctane molecules; Less energy needed to overcome the intermolecular forces; R: weaker intermolecular forces of attraction	[1] [1]
(b)(i)	 [1]: 2 proper amide linkages [1]: consists of 1 glycine, 1 alanine, 1 serine	if polymer -1m
(b)(ii)	Mr of repeat unit = 215; Number of repeat units = 600 000/215; Number of amino acids = 2790.7 x 3 = 8372.1 = 8372; R: 8373	[1] [1] [1] allow ECF

Q3

(ai)	It has the ester functional group; A: ester linkage	[1]
(aii)	Alcohol  Carboxylic acid 	[1] [1]
(b)	Add acidified potassium manganate(VII) to both solutions; If purple solution turns colourless, the sample is middle note/2-phenylethanol; If purple solution remains purple, the sample is top note/styrallyl acetate; OR Add acidified potassium dichromate(VI) solution to each sample, if the solution turned from orange to green, the sample is middle note or if the solution remained orange, the sample is top note.	[1] [1] [1]
(ci)	It has a carbon-carbon double bond;	[1]
(cii)		[1]
(ciii)	1 mol of end note reacts with 1 mol of iodine. No. of mole of end note = 100/250 = 0.4 mol; No of mole of iodine = 0.4 mol Mass of iodine =0.4 x 2 x 127 =101.6 g; R: missing unit	 [1] [1]

Q4

A6	As the number of carbon atom increases/As the molecules get larger, the boiling point	1
(a)	increases. They share the same general formula. OR The consecutive members of the group differ by $-CH_2$ OR They have the same functional group.	1
(b)	$C_nH_{2n+1}CHO$ where n = 0 for first member / $C_nH_{2n}O$ where n= 1	1
	$CH_3CH_2CH_2$ $CH_2CH_2CH_2CH_2CHO$ OR $C_7H_{15}CHO$ OR $C_8H_{16}O$	1
(c)		1
(d)	Product **A**: Propanoic acid [1]	1
	Polymer **B**: 	2
(ii)		2

Q5

(a)	The linkage present in nylon is amide linkage. The one in terylene is ester linkage.	1
	Nylon is made up of dicarboxylic acid monomers and diamine monomers while terylene	1
	is made up of dicarboxylic acid monomers and diols monomers	1
	The side products from formation of both polymers are water molecules.	
(b)	Condensation polymers are **macromolecule** made from combining many **monomers**, with an **elimination of a small molecule like water**.	1
(c)	Being non-biodegradable will allow the polymer to be durable and resistant to corrosion.	1
	However, being non-biodegradable will also create more landfills when the polymer is disposed off and thus, causing land pollution.	1
(d)	The polymer has **much higher carbon content** than its monomers thus are **less flammable** and would **undergo incomplete combustion** with oxygen to form carbon monomers and soot. In contrast, its monomers have lower carbon content and is more flammable and undergoes complete combustion to form carbon dioxide.	1 1
(e)	It is biodegradable and thus does not take up landfill space and reduce land pollution.	1
	It is carbon neutral as the same amount of carbon dioxide that is used to grow the corn plant is released upon decomposition/combustion of PLA.	1